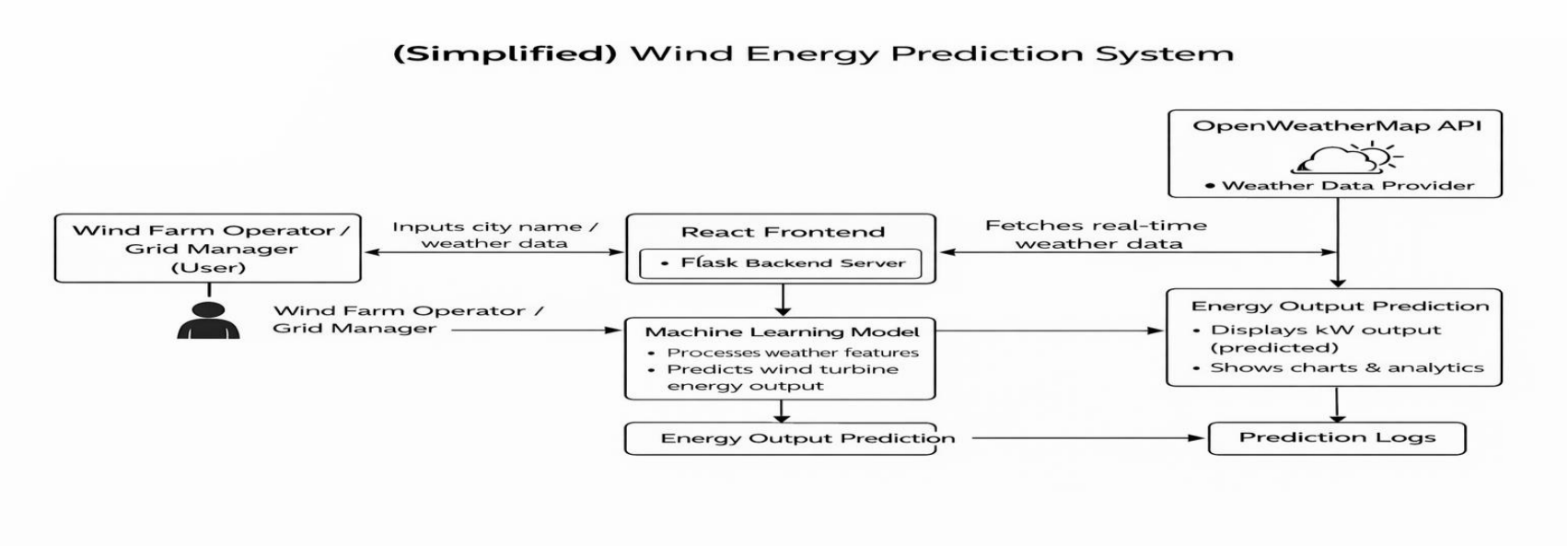
Project Design Phase-II
Data Flow Diagram & User Stories

Date	22 February 2026
Team ID	LTVIP2026TMIDS42033
Project Name	**Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management**
Maximum Marks	4 Marks

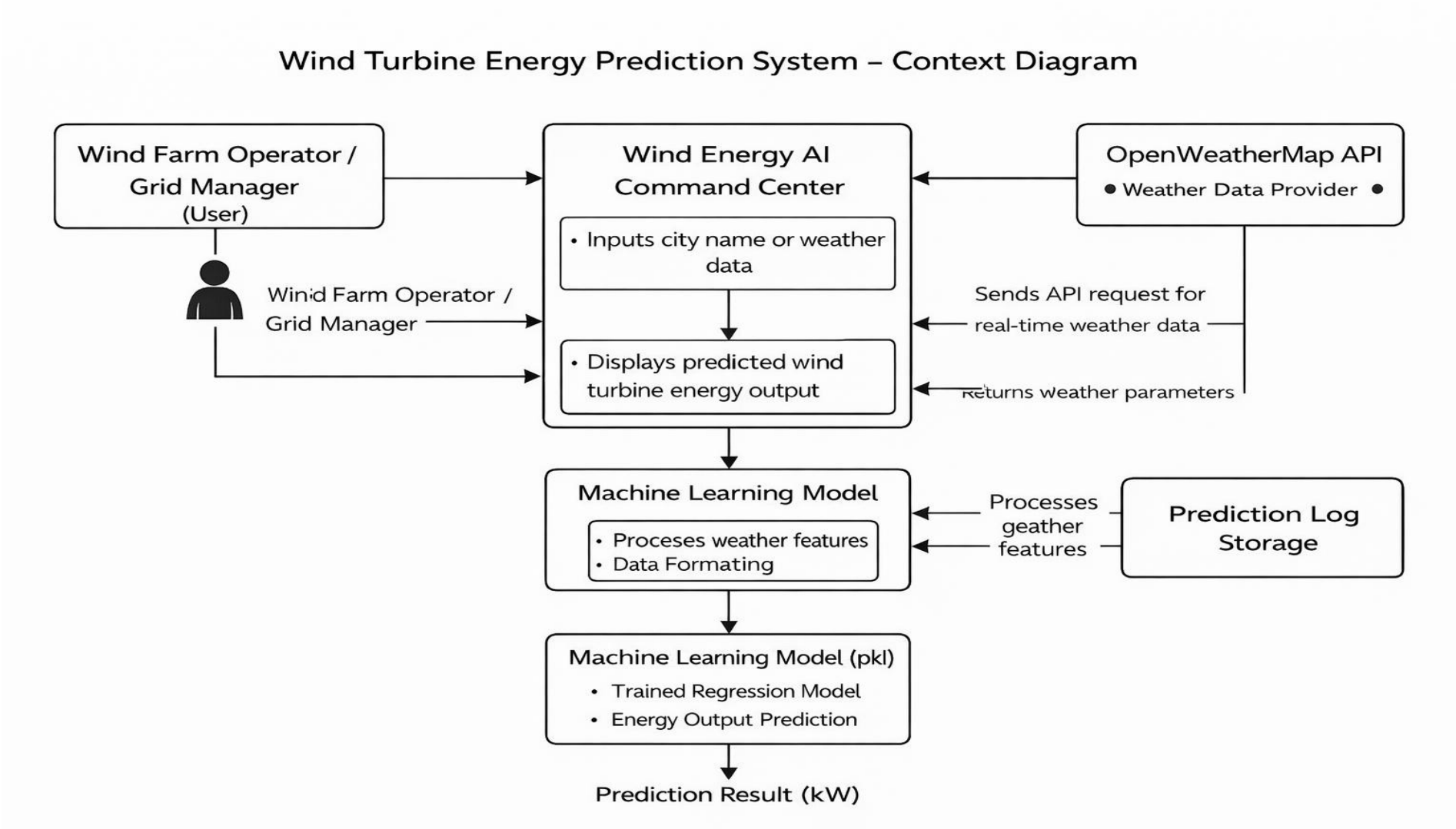
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (simplified)



Example: DFD Level 0 (Industry Standard)



User Stories

User Type	Functional Requirement	USN	User Story	Acceptance Criteria	Priority	Release
Wind Farm Operator	Weather Search	USN-1	As a user, I can enter a city name to fetch real-time weather data	Weather data displayed correctly	High	Sprint-1
Wind Farm Operator	Energy Prediction	USN-2	As a user, I can predict wind turbine energy output based on weather data	Predicted output shown in kW	High	Sprint-1
Wind Farm Operator	Visualization	USN-3	As a user, I can see energy output in charts and gauge format	Chart and gauge display correctly	High	Sprint-2
Grid Manager	Wind Direction	USN-4	As a user, I can view wind direction for grid balancing decisions	Wind direction displayed correctly	Medium	Sprint-2
Admin	Model Management	USN-5	As an admin, I can retrain the ML model with updated dataset	Model retrained successfully	Medium	Sprint-3